

2. FinOps and Cloud Economics

2.1 - FinOps as a Business and Operational Framework

FinOps provides the business framework for interpreting the project's cost data. A pipeline can reconcile charges technically while leaving allocation, ownership or business meaning unresolved. The platform therefore supplies data for financial questions as well as for reporting.

The FinOps Foundation framework defines FinOps through collaboration between engineering, finance and business functions, timely data and financial accountability. The project applies this framework to a bounded problem: preparing governed cost products and examining what decisions those products can support. It does not evaluate the maturity of the organization's full FinOps practice (FinOps Foundation, n.d.-c).

Figure 2.1 locates this study within the Framework. Data ingestion, allocation, reporting and governance are relevant to the proposed platform, but the figure does not imply that the project implements every FinOps capability.

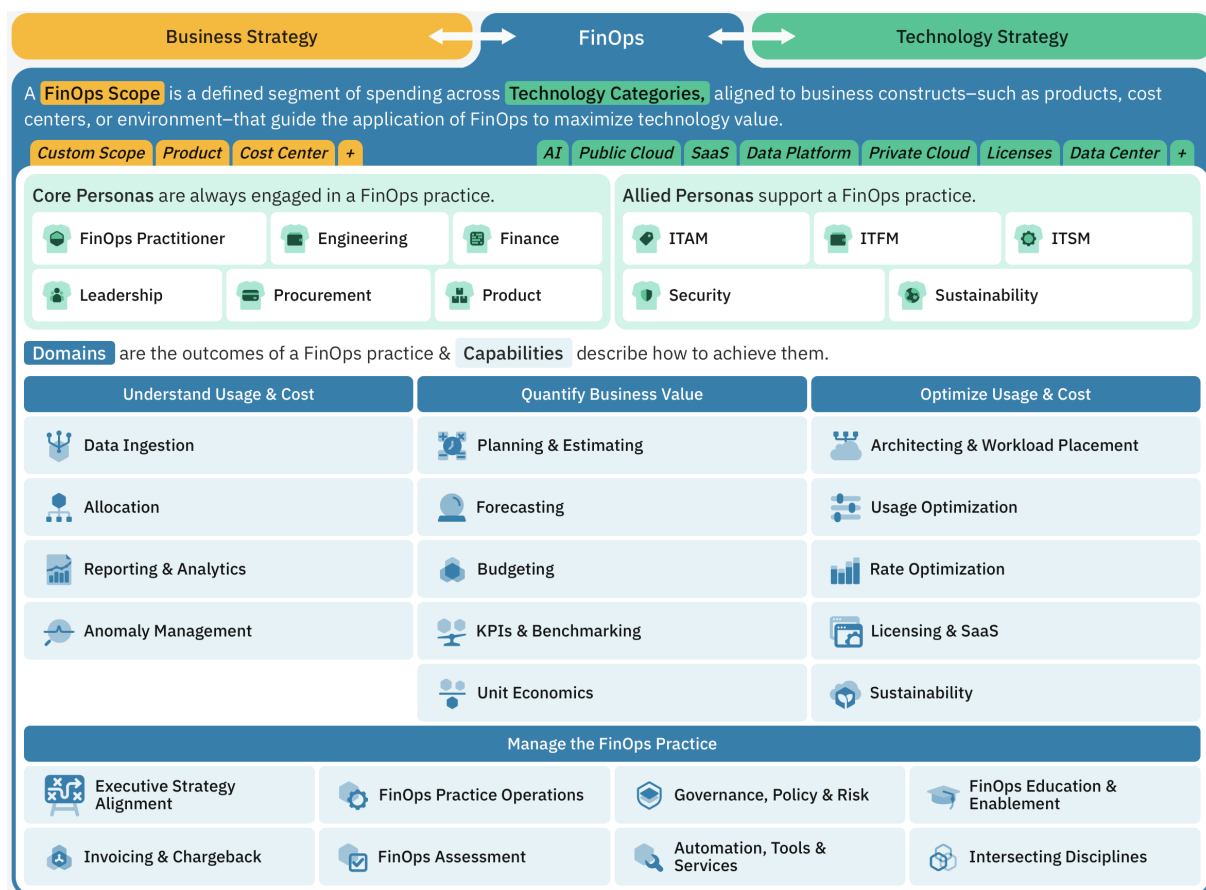


Figure 2.1 - FinOps Framework: scopes, personas, domains and capabilities. Source: FinOps Foundation, *FinOps Framework*, <https://www.finops.org/framework/> (accessed 30 September 2026). License: CC BY 4.0, <https://creativecommons.org/licenses/by/4.0/>.

Why cloud economics require FinOps

Cloud expenditure varies with usage, configuration, location and contractual terms. Resource operators, budget owners and invoice reviewers may therefore need different views of the same charge. Shared cost definitions and explicit allocation rules allow these views to be compared without treating every difference as waste or saving.

Roles in the proposed operating model

- The Data Platform function provides ingestion, transformation, quality, security and operational controls.
- FinOps owns cost semantics, allocation rules, financial indicators and business validation.
- The Power BI or dashboard function owns presentation and semantic-model standards.
- Managers and application owners consume information for their authorized perimeter.
- Authorized analysts may need direct access to governed datamarts for additional analysis.

This is a design allocation of responsibility. The existing dashboard's actual permission and change processes are described from the student's account in section 1.2; the proposed roles require organizational validation before they can be reported as established practice.

Source for this section (accessed 30 September 2026):

- FinOps Foundation, *FinOps Framework*: <https://www.finops.org/framework/>

2.2 - Inform, Optimize and Operate in the Project

Analytical positioning

The FinOps Foundation describes Inform, Optimize and Operate as recurring phases. This thesis uses them to organize the questions tested by the prototype, while recognizing that a reliable cost measure can inform several activities at once (FinOps Foundation, n.d.-e).

Inform

Inform establishes visibility and accountability. It includes cost trends, allocation, showback, comparison between periods, identification of unallocated expenditure and access to consistent definitions. In the prototype, Inform is represented principally by the validated datamarts and the Streamlit dashboard.

Optimize

Optimize evaluates whether expenditure delivers sufficient value and whether rates, commitments or resource usage can be improved. The current dataset supports comparisons among list, contracted, effective and billed cost. It does not by itself support defensible rightsizing recommendations because those require utilization, inventory, recommendation or pricing-reference data.

Operate

Operate embeds FinOps into recurring processes. For this PFE it includes ingestion, data quality, reconciliation, freshness, execution traceability and recovery. Forecasting and anomaly detection remain conditional on sufficient historical data and an explicit evaluation protocol.

Boundary of claims

The dashboard demonstrates cost visibility on synthetic data. It does not demonstrate measured organizational savings, a reduction in company workload or production-grade continuous operation. These claims require separate evidence.

Source for this section (accessed 30 September 2026):

- FinOps Foundation, *FinOps Phases*: <https://www.finops.org/framework/phases/>

2.3 - Cost Semantics and FOCUS

A common language for billing data

Cloud billing exports differ in their column names and in the meanings attached to usage, discounts and charges. The FinOps Open Cost and Usage Specification (FOCUS) defines a common structure and semantic requirements for technology cost-and-usage datasets. It gives data generators a specification for publishing billing data and data consumers a reference for interpreting it. FOCUS is neither a billing service nor a dashboard; adopting its vocabulary does not remove the need to check the contents of each export (FOCUS Project, n.d.-d).

FOCUS is an open specification initiative of the Linux Foundation, organized as a Joint Development Foundation specification project. Its steering committee approves the specification, while contributors and maintainers develop it through an open process. The FinOps Foundation provides support, but FOCUS is not a format owned by a single cloud provider (FOCUS Project, n.d.-a).

Provider support and specification versions

The FOCUS Project's provider index lists FOCUS datasets from AWS, Microsoft Azure and Google Cloud, as well as other cloud and technology vendors. On 30 September 2026, the index listed version 1.2 for those three providers. The specification itself had reached version 1.4 by that date. Provider support therefore needs to be checked by export and version; the existence of a FOCUS export does not establish that all providers publish the same release or that a particular dataset satisfies every requirement (FOCUS Project, n.d.-c; FOCUS Project, n.d.-d).

Scope of FOCUS in this study

The professional setting at Technip Energies motivates the need for consistent showback definitions, but the available evidence does not establish whether the existing company reporting flow uses a FOCUS export or which version it might use. This study does not make that claim. Its experimental input is different: two anonymized Azure FOCUS files provide reference data for a generator that produces fictional cost-and-usage records. The project's configuration identifies the target specification as FOCUS 1.0; its own data contract is versioned 1.0.0. These numbers refer to different artifacts. Microsoft documents the Azure FOCUS schema, but that documentation alone does not establish the provenance of the two reference files (Microsoft, n.d.-a).

The generated records are processed on a Databricks platform hosted on Google Cloud. That processing environment does not change the cloud provider represented by the reference billing data. The contract's Microsoft-provider and EUR-currency checks describe this experimental corpus, not universal FOCUS requirements. They also do not constitute an independent certification of full FOCUS conformance. Chapter 3 explains how the versioned contract controls data quality and future schema changes; section 2.4 examines the cost measures whose meanings the analytical model must preserve.

Sources for this section (accessed 30 September 2026):

- FOCUS Project, *About the FOCUS Project*: <https://focus.finops.org/about-focus/>
- FOCUS Project, *Get Started with FOCUS Datasets*: <https://focus.finops.org/docs/implementation/get-started/get-started-with-focus-datasets/>
- FOCUS Project, *What is FOCUS?*: <https://focus.finops.org/what-is-focus/>
- Microsoft, *FOCUS cost and usage details file schema*: <https://learn.microsoft.com/en-us/azure/cost-management-billing/dataset-schema/cost-usage-details-focus>

2.4 - From Cloud Charges to Showback

Interpreting cloud costs

A cloud charge reflects usage, a pricing quantity and unit, contractual terms, commitment benefits, credits and billing timing. The meaning of “cost” therefore depends on the measure and accounting perspective selected. FOCUS 1.0 defines four measures used in this project (FOCUS Project, 2024a, 2024b, 2024c, 2024d):

Measure	Interpretation	Use in this study
ListCost	Cost at the public list unit price	Reference for rate comparisons
ContractedCost	Cost at the negotiated contracted unit price	Comparison with list pricing where the rows are comparable
EffectiveCost	Economically allocated cost, including applicable amortized commitment benefits	Service and resource analysis on an accrual basis
BilledCost	Amount serving as the basis for invoicing in the billing period	Reconciliation and billed-cost reporting

For comparable rows with the required fields, ListCost and ContractedCost can be related to their respective unit prices and PricingQuantity. That quantity need not equal the consumed quantity. Credits, corrections, purchases and covering charges also require treatment according to their charge class. In particular, BilledCost and EffectiveCost answer different questions: a commitment purchase can be billed at one time while its economic cost is distributed over covered usage. Their difference is not automatically a saving.

Comparisons such as ListCost - ContractedCost or ListCost - EffectiveCost are analytical candidates only for a consistent currency, period, scope and charge population. They describe differences between cost perspectives, not verified business savings. A potential saving from rightsizing or a new commitment requires further evidence, including utilization, inventory, recommendations or coverage. The prototype's ListCost - EffectiveCost indicator is therefore an effective difference from list cost until its exclusions and interpretation have been validated. Chapter 6 applies these distinctions to the analytical results.

Allocating costs to business responsibility

Standardized cost columns show what was charged but do not determine which business entity should be accountable. Allocation links a charge to an application, domain, project or cost center through documented rules. Shared costs need an explicit rule; charges without a reliable owner must remain visible as unallocated. Financial reconciliation alone does not establish allocation coverage (FinOps Foundation, n.d.-b).

Showback presents attributed costs to the people responsible for them. Chargeback additionally posts or recovers those costs through an accounting process; it is a different operating choice, not an automatic next step. This project prepares showback views but does not implement enterprise chargeback (FinOps Foundation, n.d.-f).

An application owner needs the relevant application view, while a domain manager needs a consolidated domain view. The analytical model must preserve the measure and allocation rule behind each figure, including an unallocated category. Access control is a separate requirement for distributing these views safely. Whether the implemented dashboard serves both audiences remains a validation question; the data model alone does not establish it.

Sources for this section (accessed 30 September 2026):

- FOCUS Project, *FOCUS* 1.0 - *Billed* *Cost:*
<https://focus.finops.org/docs/specification/v1-0/columns/billed-cost/>
- FOCUS Project, *FOCUS* 1.0 - *Contracted* *Cost:*
<https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/contracted-cost/>
- FOCUS Project, *FOCUS* 1.0 - *Effective* *Cost:*
<https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/effective-cost/>
- FOCUS Project, *FOCUS* 1.0 - *List* *Cost:*
<https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/list-cost/>
- FinOps Foundation, *Allocation:* <https://www.finops.org/framework/capabilities/allocation/>
- FinOps Foundation, *Invoicing* & *Chargeback:*
<https://www.finops.org/framework/capabilities/invoicing-chargeback/>

2.5 - FinOps Team Activities and Outputs

The preceding sections define the FinOps concepts used in this study. An internal presentation groups the team's work into four areas and identifies Azure, Oracle Cloud Infrastructure (OCI), on-premises infrastructure and AI within its scope (Technip Energies, 2026a). Their levels of development differ. The activities below describe the team's remit; they do not measure how often each task is performed.

Measurement and accountability

- **Cost analysis and reporting.** For a requested period and perimeter, FinOps extracts and checks cost data, identifies the main drivers, and prepares a scoped analysis or dashboard view.
- **Allocation and showback.** The team examines costs by dimensions such as application, domain and tags. It reviews indicators and changes in expenditure so that stakeholders can understand their attributed costs.

Planning and forecasting

- **Cost exploration.** Before an architecture is fixed, FinOps compares possible services, providers, regions and usage assumptions. The resulting estimate is indicative.
- **Project estimation and follow-up.** Once the architecture is defined, the team estimates costs by resource and environment for review and approval. The documented process includes comparing actual costs with the estimate after deployment.
- **Budget monitoring.** Forecasts and variance analysis help the team discuss changes in expected expenditure with project and budget owners.

Cost optimization

- **Resource and architecture review.** The team examines cleanup, rightsizing and design changes as possible ways to reduce unnecessary expenditure.
- **Commitments and savings follow-up.** It reviews reservations and Savings Plans, identifies unused commitments, and tracks the effects of optimization actions.

Governance and operations

- **Coordination and controls.** FinOps participates in cost reviews, contract discussions, guardrails and action tracking with other teams.
- **Extending cost visibility.** On-premises data-centre onboarding is in progress. The presentation identifies AI cost projections and broader data-quality automation as work still under development.

This PFE addresses a narrower part of the team's work. Its platform prepares and checks synthetic, Azure-derived cost-and-usage data for analytical tables and showback. It does not approve project estimates, purchase commitments, implement accounting chargeback, or complete the team's forecasting, on-premises and AI processes.

Source for this section: Technip Energies, *FinOps Corner*, 29 September 2026. Confidential internal presentation; no public URL.

References

- FinOps Foundation. (n.d.-b). *Allocation*. <https://www.finops.org/framework/capabilities/allocation/> (accessed 30 September 2026).
- FinOps Foundation. (n.d.-c). *FinOps Framework*. <https://www.finops.org/framework/> (accessed 30 September 2026).
- FinOps Foundation. (n.d.-e). *FinOps Phases*. <https://www.finops.org/framework/phases/> (accessed 30 September 2026).
- FinOps Foundation. (n.d.-f). *Invoicing & Chargeback*. <https://www.finops.org/framework/capabilities/invoicing-chargeback/> (accessed 30 September 2026).
- FOCUS Project. (2024a). *FOCUS 1.0 - Billed Cost*. <https://focus.finops.org/docs/specification/v1-0/columns/billed-cost/> (accessed 30 September 2026).
- FOCUS Project. (2024b). *FOCUS 1.0 - Contracted Cost*. <https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/contracted-cost/> (accessed 30 September 2026).
- FOCUS Project. (2024c). *FOCUS 1.0 - Effective Cost*. <https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/effective-cost/> (accessed 30 September 2026).
- FOCUS Project. (2024d). *FOCUS 1.0 - List Cost*. <https://focus.finops.org/docs/specification/v1-0/columns/cost-and-usage/list-cost/> (accessed 30 September 2026).
- FOCUS Project. (n.d.-a). *About the FOCUS Project*. <https://focus.finops.org/about-focus/> (accessed 30 September 2026).
- FOCUS Project. (n.d.-b). *FOCUS: FinOps Open Cost and Usage Specification*. <https://focus.finops.org/> (accessed 29 September 2026).
- FOCUS Project. (n.d.-c). *Get Started with FOCUS Datasets*. <https://focus.finops.org/docs/implementation/get-started/get-started-with-focus-datasets/> (accessed 30 September 2026).
- FOCUS Project. (n.d.-d). *What is FOCUS?* <https://focus.finops.org/what-is-focus/> (accessed 30 September 2026).
- Microsoft. (n.d.-a). *FOCUS cost and usage details file schema*. <https://learn.microsoft.com/en-us/azure/cost-management-billing/dataset-schema/cost-usage-details-focus> (accessed 30 September 2026).
- Technip Energies. (2026a, 29 September). *FinOps Corner* [Confidential internal presentation; not publicly accessible].